



United International University

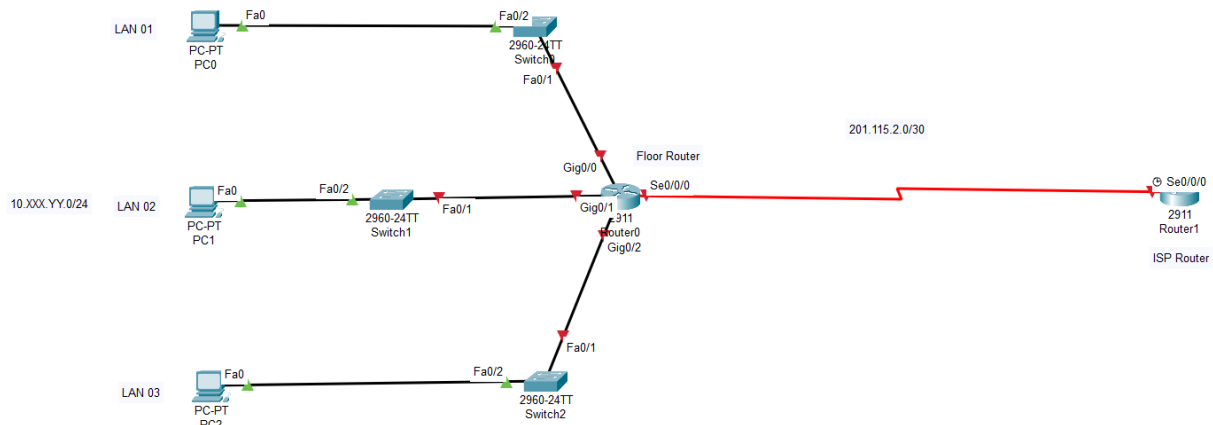
Department of Computer Science & Engineering

CSE 3712: Computer Networks Lab

Project

Total Marks: 15 + 10

Topology



IP Addressing Information

XXX = Last three digits of your Student ID % 255
YY = Last two digits of your Student ID

You need to divide the network **10.XXX.YY.0/24** into subnets such that each subnet can support at least 50 hosts.

Network Segment	Network Address
LAN 01	First Subnet
LAN 02	Second Subnet
LAN 03	Third Subnet
Floor Router – ISP Router	201.115.2.0/30

Important Notes

- Floor Router acts as: DHCP server for both LANs and NAT (PAT) router
- PCs of LAN 01 and 02 must obtain IP addresses automatically (DHCP)
- PCs of LAN 03 will obtain IP addresses manually (static)

Tasks

A sample topology file is attached with this assignment. Complete the following tasks on that sample file.

Task 01: Interface Configuration

Configure the interfaces on Router0 (Floor Router) and Router1 (ISP Router) according to the topology diagram.

1. Assign the first usable IP address of each respective subnet to Router0's GigabitEthernet interfaces (Gig0/0, Gig0/1, and Gig0/2) to act as the default gateways for the LANs.
2. Configure the Serial interfaces connecting Router0 (Se0/0/0) and Router1 (Se0/0/0) using the WAN IP addresses provided above. Ensure the links are turned on.
3. Set Router0 and Router1 names as StudentID-R0 and StudentID-R1 respectively.

Verification: Ensure all physical links show green arrows in Packet Tracer.

Task 02: DHCP Configuration

Router0 (Floor Router) will act as the DHCP server for LAN 01 and LAN 02 networks. LAN 03 will use static IP addressing.

1. Create necessary DHCP pools for the LAN 01 and LAN 02 network.
2. Configure both pools to assign the correct Default Gateway to the PCs.

Verification: Set PC0 and PC1 to DHCP and verify they successfully pull an IP address, Subnet Mask, and Default Gateway. Configure PC2 with a static IP from its respective subnet.

Task 03: Default Routing

To ensure packets going to Router1 (ISP Router), configure default routing at Router0 (Floor Router) using the following command:

```
StudentID-R0(config)# ip route 0.0.0.0 0.0.0.0 <IP of Se0/0/0 interface of Router1>
```

This ensures all the packets travel to the interface Se0/0/0 of Router1.

Task 04: Security and Access Control Lists (ACLs)

Implement security policies using standard ACLs to restrict unauthorized access across the network.

Requirement A: Protect LAN 03 (Internal Security)

LAN 03 is a highly secure department. Configure and apply an ACL such that no traffic from LAN 01 or LAN 02 is allowed to enter LAN 03.

Requirement B: Protect the Private Network (External Security)

Configure and apply an ACL to ensure that no traffic originating from the outside network (Router1) can initiate a connection into the private networks (LAN 01, LAN 02, LAN 03).

Task 05: Implement Port Address Translation (PAT)

The internal LANs represent a private corporate network. You must configure PAT (NAT Overload) on the Floor Router (Router0) so internal PCs can communicate with the outside network (Router1).

1. Define the appropriate NAT Inside interfaces (Gig0/0, Gig0/1, Gig0/2) and the NAT Outside interface (Se0/0/0).
2. Apply the NAT overload command to translate the private IP addresses to the public IP address of the Serial0/0/0 interface.

Verification

- Test connectivity using Simulation Mode:
 - PC0 → PC1 (should reach)
 - PC0 → PC2 (should not reach)
 - Any PC → ISP router (should reach)
 - ISP → Any PC (should not reach)

Submission & Marking

- A sample pkt file is uploaded. Do not change any interfaces.
- Modify the file and submit your completed .pkt (Packet Tracer) file.
This will carry 15 marks.
- Deadline: Before Lab 11 **(07 June 2026)**.
- Your submission will be checked during Lab 11 and you will be asked some questions or about how you completed the task.
This will carry 10 marks.
- You can take help from your friends, AI agents etc. However, you should understand your work.